

Garv Arora

Computer Science Engineer | Intelligent Robotic Systems & Software

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Education

Vellore Institute of Technology (VIT)

Bachelor of Technology – Computer Science & Engineering (Specialization: IoT) | CGPA: 8.36 / 10.0

Vellore, India

Aug 2023 – Jul 2027

Gems International School (CBSE)

Class XII: 76.4% | Class X: 89.3%

Gurugram, India

Nov 2016 – Apr 2023

Work Experience

LG Soft India (LGSi) – HS/ES India Lab

Living Solution Control Development Intern

- Developed C/C++ embedded control logic and state machine handlers for living solution control modules, validating real-time execution on target microcontrollers; certified as “Technically Competent” by LGSi engineering management.

Noida, India

Jun 2026 – Jul 2026

Samsung PRISM

Data Intelligence Agent Developer (Internship)

- Engineered a multi-agent data intelligence pipeline using LangGraph, FastAPI, Redis, and Qdrant to automate dataset discovery, PII redaction (spaCy, Presidio), and licensing compliance checks across HuggingFace corpora.

Remote

Apr 2026 – Oct 2026

Wissen Baum Engineering Solutions LLP

Software Automation Intern

- Built a Python BDD test automation framework using Gherkin, pytest, and Playwright, parallelizing GitLab CI pipelines to cut build times from 10 to 6 minutes (40% speedup) and reducing manual testing effort by 80%.

Pune, India

May 2025 – Jul 2025

SEDS India (SEDS VIT)

R&D Lead & Autonomous Systems Developer

- Architected a ROS2 Nav2 and RTAB-Map 3D Visual SLAM pipeline on an Intel RealSense D455 depth camera, generating elevation maps that reduced manual rover teleoperation time by 70% during competition field runs.
- Optimized computer vision pipelines using CUDA GPU kernels, YOLOv8 object detection, and Linux V4L2 zero-copy buffer management, cutting video latency by 60% across 4 simultaneous camera streams.
- Flashed Micro-ROS firmware onto ESP32 microcontrollers, combining wheel encoder and IMU telemetry with an Extended Kalman Filter (EKF) to cut odometry drift by 40% (sub-10 cm positioning accuracy).
- Shipped a real-time PyQt5 mission dashboard for telemetry visualization, motor control, and autonomous state machine sequencing during live planetary rover competition runs.

Vellore, India

Apr 2024 – Jul 2026

Projects

teleop-cursor | ROS 2, Python, rclpy, geometry_msgs/Twist | pypi.org/project/teleop-cursor

Feb 2026 – Apr 2026

- Developed a ROS 2 mouse cursor teleoperation package that converts screen displacement into geometry_msgs/Twist velocity commands at 10 Hz for Gazebo robotic simulation.

HayaiOS – Preemptive Bare-Metal RTOS Kernel | C, ARM Assembly, ARM Cortex-M4

Jan 2026 – Mar 2026

- Implemented a preemptive bare-metal RTOS kernel in C and ARM Assembly for Cortex-M4 processors, featuring a 1 ms tick scheduler, mutex/semaphore primitives, and hardware abstraction layer (HAL).

Captivity CLI | Rust, Python, Linux D-Bus, systemd, Socket IPC | pypi.org/project/captivity-cli

Nov 2025 – Jan 2026

- Built a lightweight WiFi captive portal daemon in Rust and Python with systemd background integration, D-Bus NetworkManager monitoring, keyring credential encryption, and <50ms HTTP login probing.

3D Reconstruction – IMU-Enhanced KinectFusion | C++, OpenCV, Intel RealSense SDK

Aug 2025 – Oct 2025

- Stabilized KinectFusion TSDF reconstruction by fusing RealSense D455 depth data with 6-DOF IMU telemetry via complementary filtering, reducing tracking failures by 50% during fast motion.

Patents

Patent Published (App IN202641072249 A1): Autonomous Radar Survivor Detection

Jun 2026

- Fused 24GHz FMCW mmWave radar, ultrasonic array, and IMU dead-reckoning on dual-core ESP32; formulated confidence grid decay $C = \max(0, C - \beta \Delta t)$ for survivor breathing detection under rubble.

Achievements & Awards

Amazon ML Summer School 2026: Selected among Top 3,000 out of 1,30,000+ applicants across India (~2.3% acceptance rate).

International Rover Challenge (IRC): Ranked 13th Globally (2025) and 17th Globally (2026) with Team Vyadh (SEDS VIT).

Certifications

AWS Certifications: AWS Certified AI Practitioner | AWS Certified Cloud Practitioner

Oracle Cloud Certifications: Certified Data Science Professional | Certified Generative AI Professional

Technical Skills

Languages: C, C++, Python, Rust, ARM Assembly, Java, JavaScript, SQL, Bash, Verilog

Robotics & Autonomous Systems: ROS2 (Nav2, MoveIt2), RTAB-Map (3D SLAM), Micro-ROS, EKF/UKF, PID/MPC, Sensor Fusion, mmWave Radar

Perception & Machine Learning: OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, Transformers

Embedded & Systems: ARM Cortex-M, ESP32, FreeRTOS, I2C/SPI/UART, Bare-Metal C/Assembly, Linux Kernel/D-Bus, Docker, Git, CI/CD, AWS